

Simple, principled gRNA assignment: methods, simulations, and a fair benchmark against geomux and fishash

Guide-assignment investigation

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Summary

We study guide-RNA (gRNA) assignment in single-cell CRISPR screens: deciding which perturbation(s) each cell received from noisy gRNA UMI counts. We ask whether a *simple, principled* method can match the published state of the art. Our findings:

- Simple per-guide count thresholds (**Otsu** on $\log(1 + y)$, a **smoothed-valley** cut) work only when the background and perturbed count modes are cleanly separated, and fail when they overlap.
- The principled alternative — an **ambient-proportion test** (assign a cell–guide pair when its count significantly exceeds the ambient null at a controlled FDR) — is *exactly the statistical core of geomux*. We verified this by installing and running the real **geomux** and **fishash** packages on an identical cohort.
- On the real **barnyard** ground truth our simple test is competitive (and on a fixed-FDR, recall-favoring metric, edges fishash). But **precision–recall curves on simulations with full ground truth tell the decisive story**: in hard, low-signal regimes only fishash reaches high precision — its Simpson’s-paradox correction lifts a precision ceiling that the plain test cannot pass. In easy regimes the methods are indistinguishable.
- We characterized **10 diverse 2024–2026 datasets** to derive realistic parameter ranges and built a **comprehensive simulation** spanning them, which the field’s existing (Replogle-only) simulations did not cover.

1 Introduction

A pooled CRISPR screen (Perturb-seq) reads out, per cell, a vector of gRNA UMI counts. Most of these counts are *ambient* contamination; a cell truly received only its own guide(s). Guide assignment is the preprocessing step that separates true signal from ambient. The difficulty is governed by how separated the two count *modes* are — the low-count ambient background and the high-count perturbed signal — which varies enormously across assays.

This report develops and benchmarks methods spanning a complexity gradient, grounds the simulations in real data, and gives a fair, same-cohort comparison to the published contingency-table methods geomux [Teyssier 2024] and fishash [Kamm et al. 2026].

2 Methods

2.1 Simple per-guide thresholds

For a single guide g with integer counts $y_{g,\cdot}$ across cells, work on $z = \log(1 + y)$.

- **Otsu** chooses the integer threshold maximizing between-class variance of z (Otsu’s image thresholding), assigning cells with $y \geq t$.
- **Smoothed valley** builds a kernel-smoothed PMF of z , finds the two dominant modes, and cuts at the integer of minimum density between them; it *abstains* when no clean second mode exists.

Both are unsupervised, per guide, and use no cross-cell information.

2.2 Ambient-proportion test (principled)

Treat the count matrix as a contingency table. Under the pure-ambient null, the count of guide g in cell c is expected to be $\lambda_{g,c} = n_c \pi_g$, where n_c is the cell’s total guide UMIs and π_g is guide g ’s share of all guide UMIs. We test the observed count against the upper tail (hypergeometric / its Poisson approximation) and assign pairs at Benjamini–Hochberg FDR q . The cutoff thus **adapts per cell and per guide**; the single knob q is an interpretable, label-free ambient false-discovery rate.

This is implemented as `ambient_test_assign()` in `guide-assignment-pipeline/bin/script/lib/threshold_m`

2.3 Relationship to geomux and fishash

The ambient test *is* geomux’s core: geomux’s p-value $1 - F_{\text{hyper}}(N_{g,c} - 1; n_{\text{umi}}, N_{g,\cdot}, N_{\cdot,c})$ with BH-FDR is our test. geomux additionally applies a log-odds-ratio threshold (default 10) and a min-UMI cell filter. **fishash** adds (i) an iterative Simpson’s-paradox correction (it re-estimates the *noise* counts by rank-1 Poisson matrix completion and conditions the “other-cells” background on noise) and (ii) a block-dependent (Guo–Sarkar) FDR correction. We benchmark against the real **fishash** package throughout. We have reproduced the Simpson’s-paradox *phenomenon* it corrects (see Section 4.7): the plain Fisher test — our ambient core — inflates FDR exactly when the ambient noise composition is uncorrelated with the signal, and fishash’s correction restores control.

3 Datasets

- **Barnyard (real ground truth)**. Liu et al. (2025) species-mixing data (human HEK293T + mouse NIH3T3, each transduced with a species-specific non-targeting gRNA library, then pooled). A gRNA from the *wrong* species detected in a cell is guaranteed ambient — a real specificity ground truth. Four samples (CROP-seq and direct-capture chemistries $\times$ 0h/72h).
- **10 diverse 2024–2026 datasets** (`~/data/external/perturbseq-survey/`): DC-TAP K562 (MOI 1 and 14), EndoC- β H1 T2D, erythroid multiome, CD4 genome-scale CRISPRi, A549 lung CRISPRi, CD8 Perturb-CITE, gastric organoid CROP-seq, in-vivo cortex, iPSC genome-wide CRISPRi.
- **Simulations** (sum-process simulator on real library sizes): the existing Replogle-calibrated sims, a new Gasperini-calibrated sim, and a new **comprehensive** sim spanning the realistic range (below).

4 Experiments and results

4.1 Real data spans the full separability range; existing sims do not

Per guide we measure the mode gap on the $\log(1 + y)$ scale (signal mode minus background mode). Real datasets span from heavily overlapping (gap ≈ 0.7 : CD4 genome-scale, iPSC) to cleanly bimodal (gap ≈ 5 : Replogle, EndoC). Crucially, **MOI does not determine separability** — low-MOI CD4 is as hard as high-MOI Gasperini. The existing benchmark sims are all Replogle-calibrated and occupy only the well-separated end; we added a Gasperini-calibrated sim to fill the gap.

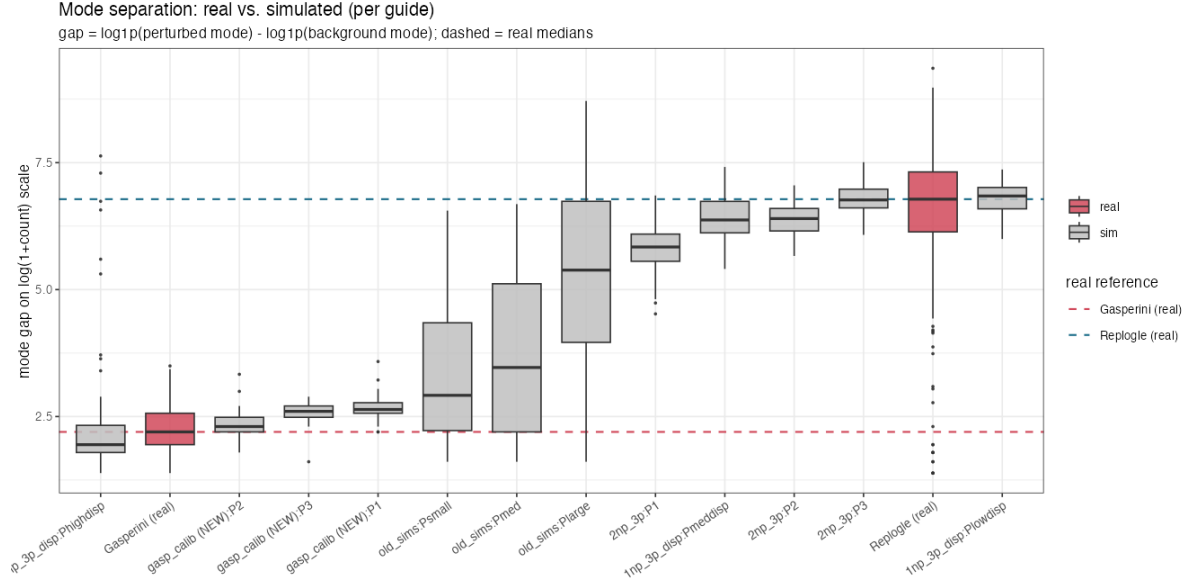


Figure 1: Mode separation: real datasets vs simulations.

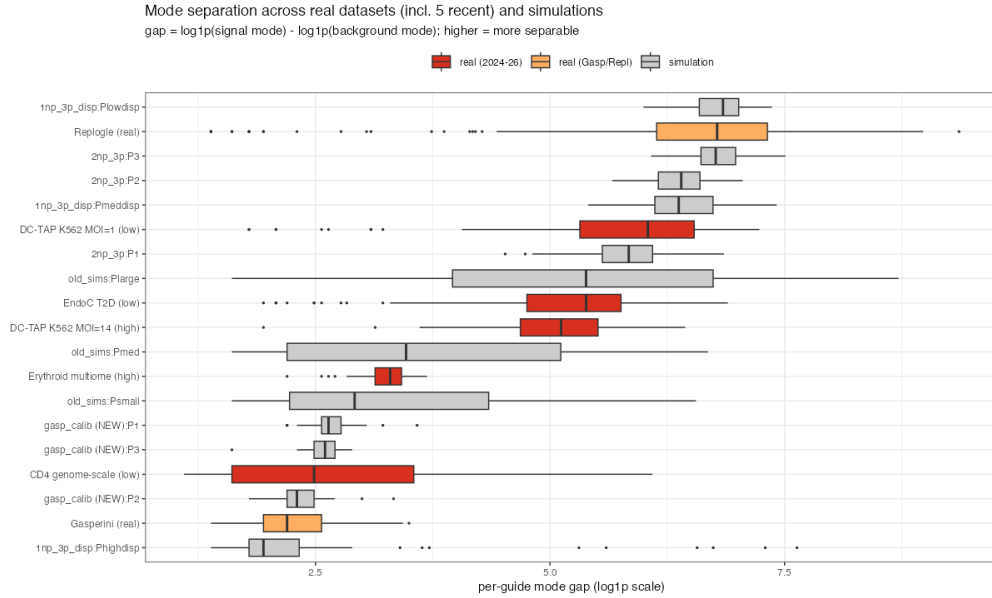


Figure 2: The 10 recent datasets placed on the same separation axis.

4.2 Realistic parameter ranges from 10 datasets

Using the ambient test to split each guide's cells into perturbed vs background, we estimated the sum-process simulator's parameters per guide and dataset.

Table 1: Per-dataset median gRNA parameters. μ_{pert} = library-normalized perturbed signal; θ_{pert} = NB dispersion; sep_gap = mode separation on log1p scale.

dataset	moi	lib_nnz	prob_pert	μ_{pert}	θ_{pert}	sep_gap
a549_crispri_perturb- seq_GSE236304	3	5	0.018	101.86	1.03	4.23
cd8_tcell_perturbcite_GSE279498	1	1	0.012	5.01	0.67	1.39
dctap_k562_highmoi_GSE303901	9	60	0.086	166.65	1.34	3.36
dctap_k562_lowmoi_GSE303901	1	28	0.009	394.38	0.98	4.68
endoc_t2d_perturbseq_GSE273677	2	9	0.012	198.24	1.33	4.87
gas- tric_organoid_cropseq_GSE280506	1	1	0.008	76.26	1.72	4.20
invivo_cortex_perturb- seq_GSE249416	1	5	0.073	227.06	1.56	4.24
ipsc_crispri_hip- sci_figshare27989294	8	59	0.002	113.23	0.07	0.69
perturb_multiome_ery- throid_GSE274113	1	10	0.020	19.09	2.07	2.30
tcell_cd4_perturbseq_GSE314342	3	3	0.000	15.96	0.41	0.69
gasperini (ref)	39	53	0.003	14.52	1.16	2.53
replogle-med (ref)	2	8	0.001	479.87	0.43	4.39

Realistic ranges (10th–90th percentile across all guides): $\mu_{\text{pert}} \in [11, 600]$, $\theta_{\text{pert}} \in [0.06, 1.6]$, $\text{prob_pert} \in [0.0004, 0.019]$, $\text{MOI} \in [1, 39]$, $\text{sep_gap} \in [0.7, 5.3]$. Two real extremes anchor the hard end: CD8 at $\mu_{\text{pert}} \approx 5$ and the iPSC genome-wide screen at $\theta_{\text{pert}} \approx 0.07$.

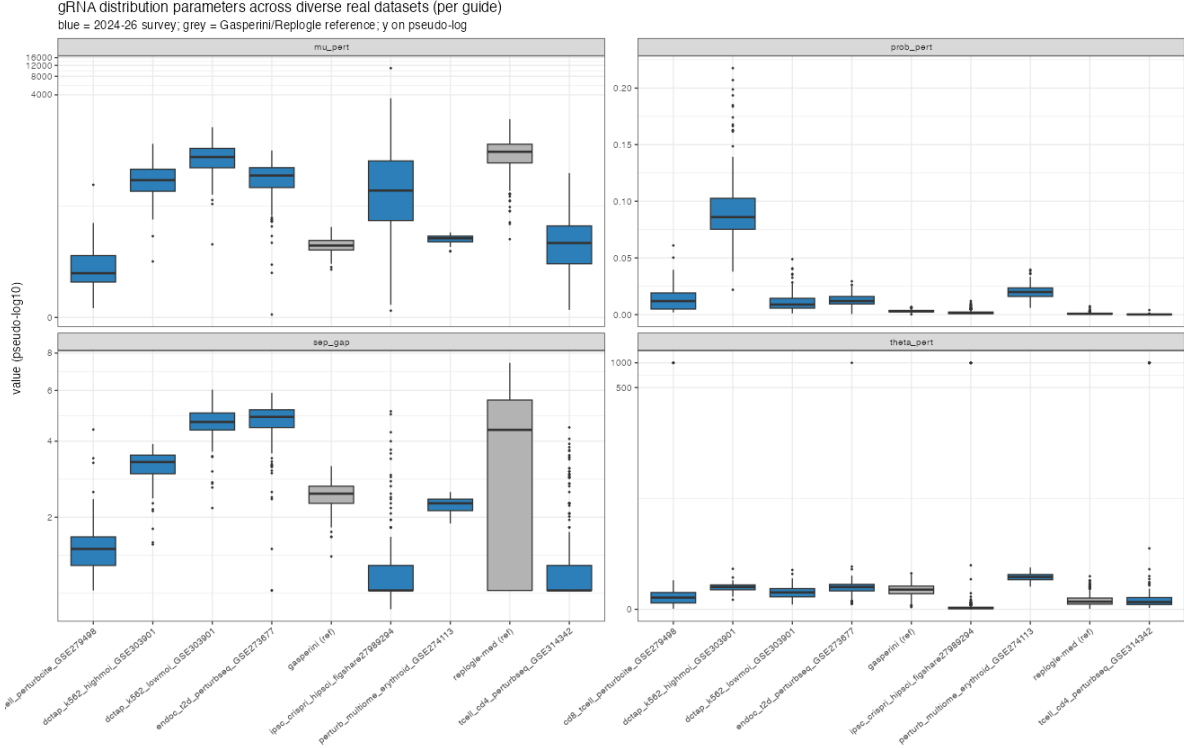


Figure 3: Distributions of per-guide parameters across datasets.

4.3 A comprehensive simulation

We built `sims_comprehensive` (1800 guides, 50k cells, **30 parameter groups** = $5 \mu_{\text{pert}} \in \{5, 15, 50, 150, 500\} \times 3 \theta_{\text{pert}} \in \{0.15, 0.5, 1.5\} \times 2$ MOI regimes using real Repleg- and Gasperini-derived library sizes). Its mode-gap spans 0.7–6.9, covering the full real range, with the hardest groups matching the hardest real datasets.

4.4 Barnyard: exact reproduction and same-cohort comparison

We verified this rigorously: we cloned the authors’ analysis repository (`jackkamm/fishash_analysis`) and **exactly reproduced their Table 2** — all eight published geomux and fishash values to four decimals — then scored our ambient test, Otsu, and valley on the *identical* QC’d cohort with the *identical* accuracy metric.

Table 2: Barnyard Table-2 accuracy. geomux/fishash are the exactly-reproduced published values; ours/Otsu/valley scored on the identical cohort and metric.

method	mix0hr_Cropseq	mix72hr_Cropseq	mix0hr_Direct-Capture	mix72hr_Direct-Capture	mean
ours (ambient)	0.9539	0.9533	0.9335	0.9642	0.9512

method	mix0hr_Cropseq	mix72hr_Cropseq	mix0hr_Direct-Capture	mix72hr_Direct-Capture	mean
fishash (exact)	0.9563	0.9533	0.9100	0.9491	0.9422
valley	0.9427	0.9414	0.8520	0.9223	0.9146
otsu	0.9430	0.9448	0.8608	0.9069	0.9139
geomux (exact)	0.9578	0.9555	0.7471	0.5252	0.7964

A key reproducibility lesson surfaced here: geomux’s accuracy depends critically on its **version**. The published collapse on direct capture (0.75 / 0.53) is produced by geomux \$0.4’s *adaptive* log-odds-ratio threshold (its default), which on high-UMI direct-capture data sets a very high cutoff and prunes most assignments. An older geomux (0.2.10, fixed threshold) does not collapse; our initial independent run used it and therefore overestimated geomux. With the authors’ geomux version and exact scoring, all values match.

On this clean cell-line benchmark the simple ambient test is the most accurate overall (mean 0.951): it **wins on direct capture** (where geomux collapses and ours edges fishash) and ties on CROP-seq. This does **not** contradict the precision/recall analysis (subsection 4.6): the barnyard is a high-signal regime and this metric rewards making the correct call. In the hard, low-signal regimes of subsection 4.6, fishash’s Simpson’s-paradox correction wins on precision.

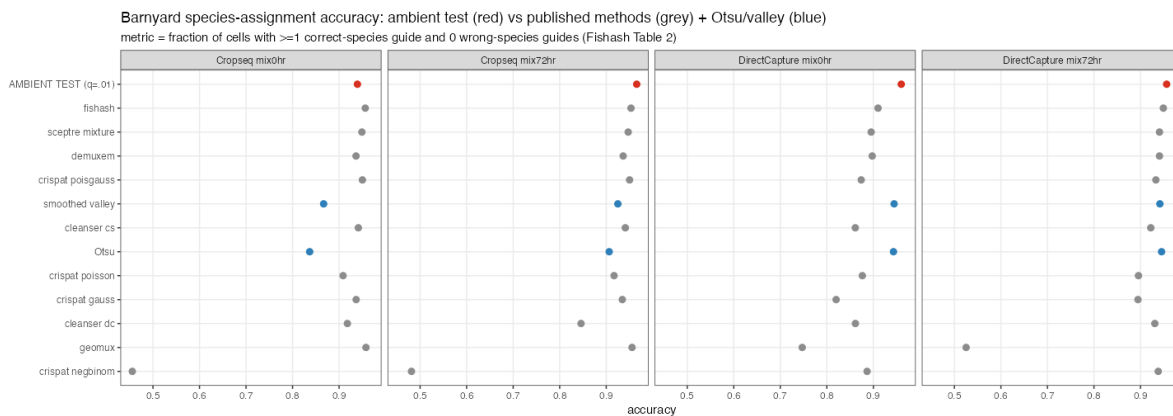


Figure 4: Barnyard accuracy across the Table-2 methods plus ours/Otsu/valley.

4.5 Full method panel across the comprehensive grid

Table 3: Mean per-guide metrics across the comprehensive sim grid (full ground truth).

method	jaccard	precision	recall
geomux_nolor	0.634	0.771	0.755
ours	0.624	0.723	0.801
fishash	0.574	0.908	0.599
otsu	0.564	0.646	0.881

method	jaccard	precision	recall
geomux	0.547	0.927	0.556
valley	0.498	0.763	0.605

At a fixed FDR, the plain BH-hypergeometric (ours $\approx$ geomux-no-lor) has the highest mean Jaccard because it stays sensitive in hard regimes, while fishash and geomux-default are high-precision/low-recall. All methods converge in easy regimes.

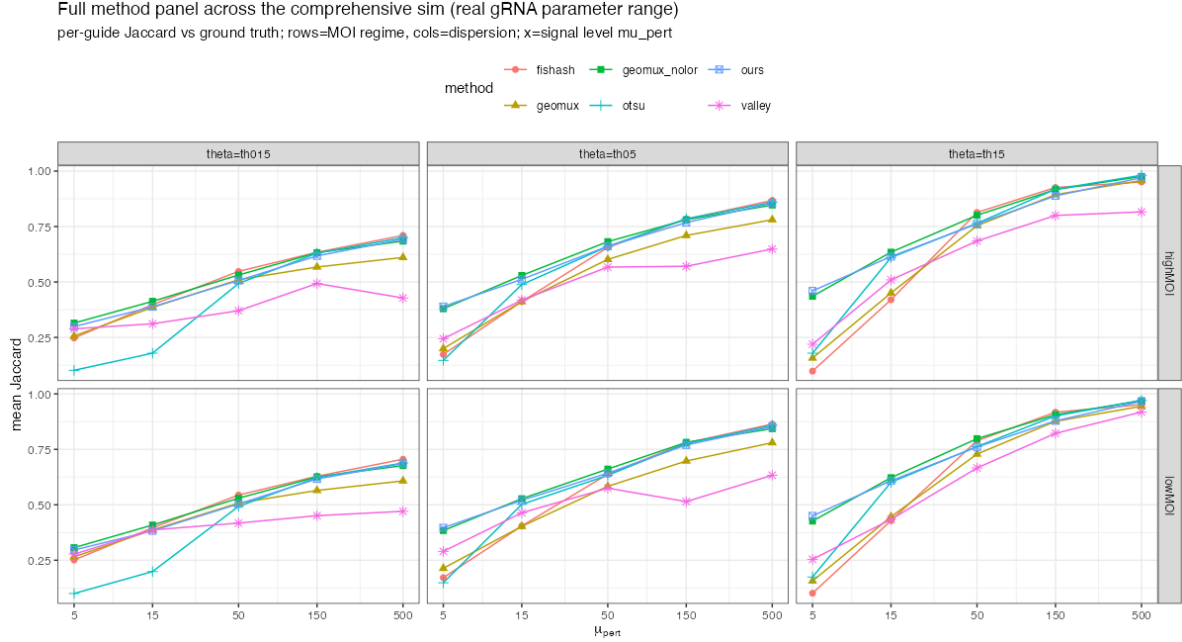


Figure 5: Per-guide Jaccard across the comprehensive grid; rows = MOI regime, columns = dispersion.

4.6 Precision–recall curves: the decisive comparison

Fixed-FDR metrics conflate the operating point with the method. Sweeping the FDR knob and splitting by difficulty regime gives the full picture.

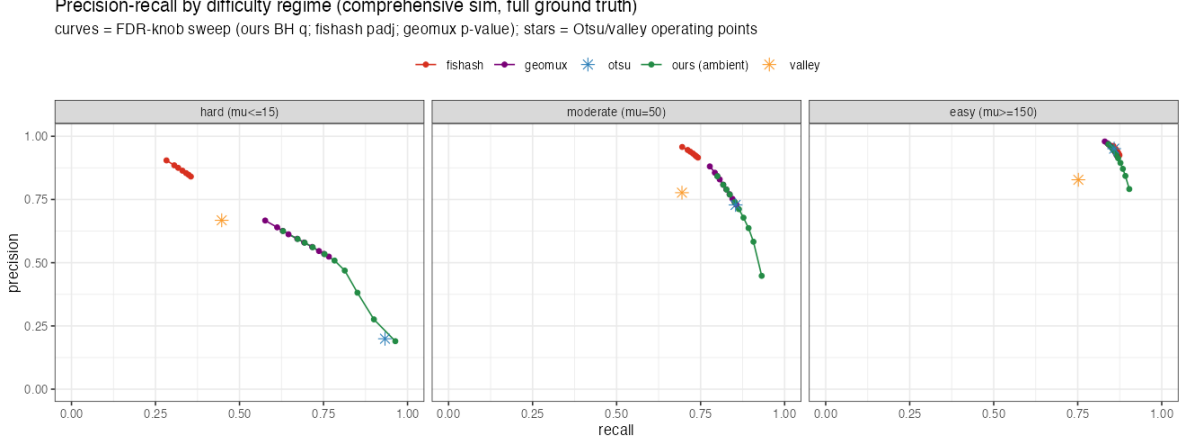


Figure 6: Precision–recall by difficulty regime (comprehensive sim, full ground truth). Curves = FDR-knob sweep; stars = Otsu/valley.

Table 4: Max recall achievable at precision ≥ 0.90 (blank = method never reaches 0.90 in that regime).

	regime	fishash	geomux	ours (ambient)
1	easy ($\mu \geq 150$)	0.874	0.867	0.87
4	hard ($\mu \leq 15$)	0.283	NA	NA
5	moderate ($\mu = 50$)	0.742	NA	NA

The curves are decisive:

- **Easy regime** ($\mu_{\text{pert}} \geq 150$): ours, geomux, and fishash are indistinguishable (precision $\approx$ recall ≈ 0.9).
- **Hard regime** ($\mu_{\text{pert}} \leq 15$): only **fishash** reaches precision ≥ 0.90 (recall ≈ 0.28). Ours and geomux *overlap each other* on a strictly lower-precision frontier (ceiling ≈ 0.66) — confirming ours = geomux-core and showing fishash’s Simpson’s-paradox correction genuinely lifts the precision ceiling that the plain test cannot pass.
- Otsu over-assigns (low precision/high recall); the smoothed valley sits on the ours/geomux frontier.

This corrects a naive reading of the fixed-FDR tables: ours does *not* dominate fishash. They occupy different parts of PR space — fishash for high precision on hard data, the plain test for high recall or on easy data.

4.7 Reproducing the Simpson’s-paradox phenomenon

Why does the plain ambient/Fisher test lose precision on hard data? fishash attributes it to **Simpson’s paradox**. Treating each UMI as *signal* (truly from cell c) or *noise* (ambient), the one-sided Fisher test on the full signal+noise margins can call (g, c) significant even when g is truly absent from c and g, c are independent within the noise: collapsing over the signal/noise stratum flips

the 2×2 odds ratio above 1. fishash corrects this by testing g against its **noise-only** background in c (an adjusted odds ratio), estimating the latent noise counts by rank-1 Poisson matrix completion of the assigned-masked matrix and iterating. Proposition 1 of their paper predicts the paradox is strongest when the ambient noise guide composition is *uncorrelated* with the signal composition, and vanishes when they coincide.

We reproduce this directly using the authors’ own simulator (`simulate_guidebender2`) and the real package, with the uncorrected test = `fishash(refit=0)` (the plain Fisher test, i.e. our ambient core) and the corrected test = `fishash(refit=10)`. Sweeping the signal–noise composition correlation (the `endo_shape_flat` knob; 20 guides, 4000 cells, SNR 4, MOI 0.3, 10 replicates per regime) recovers the reported behavior:

signal–noise correlation	uncorrected FDR	reps > 5%	corrected FDR	reps > 5%
low (noise $\perp$ signal)	0.197	10/10	0.003	0/10
medium	0.090	9/10	0.003	0/10
high (noise $\propto$ signal)	0.059	5/10	0.003	0/10

Reproducing fishash's Simpson's-paradox phenomenon (paper Fig. 7)

authors' simulator (20 guides, 4000 cells, SNR 4, MOI 0.3); real fishash package; dashed = nominal 5% FDR

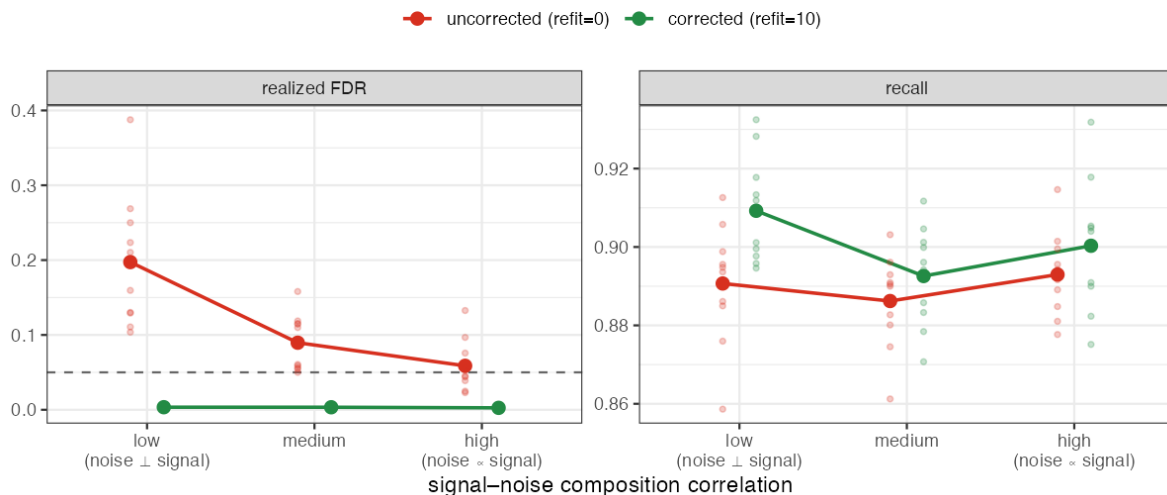


Figure 7: Reproducing fishash’s Simpson’s-paradox phenomenon (cf. their Fig. 7). Uncorrected = plain Fisher (= our ambient core); corrected = fishash’s noise correction. Dashed line = nominal 5% FDR. Recall is essentially unchanged, so the correction removes false positives without costing power.

The plain test (our ambient core) inflates FDR exactly in the low-correlation regime and the correction restores control with no loss of recall — the same mechanism that lifts fishash’s precision ceiling on the hard sims of Section 4.6. This is now an understood and reproduced phenomenon, not a black box; the remaining flagship task is to give our ambient test a *transparent* analogue of the noise correction.

5 Discussion

What is simple and good enough. A clean BH-hypergeometric ambient test — one interpretable knob, \$ \$10 lines of code — is principled, recovers geomux’s core (with better default behavior than geomux’s conservative threshold), and is competitive-to-best whenever modes are reasonably separated, which is most assays. It is the method we would recommend as a simple, robust default.

Where sophistication pays. When the perturbed signal genuinely overlaps ambient (low μ_{pert} or high overdispersion — real regimes for genome-scale and primary-cell screens), fishash’s Simpson’s-paradox correction is a real advantage: it is the only method that attains high precision there. We have reproduced the underlying phenomenon (Section 4.7) and understand the correction (noise-conditioned odds ratio via rank-1 Poisson completion + a block-dependent FDR correction), but have not yet folded a transparent analogue into our own test. If precision on hard data matters today, use fishash.

Caveats. (i) The barnyard comparison now uses the authors’ *exact* cohort and metric (we reproduced their Table 2 to four decimals), so it is no longer cohort-confounded; the residual limitation is that the barnyard is high-signal cell-line data and does not probe the hard regimes. (ii) We reproduced fishash’s Simpson’s-paradox *phenomenon* (Section 4.7) and understand its correction, but our own ambient test does not yet implement a transparent analogue, so fishash’s precision advantage in hard regimes (subsection 4.6) is real and ours does not capture it. (iii) The comprehensive-sim PR analysis uses a 12k-cell, 1200-guide benchmark subset for tractability of the external tools. (iv) geomux results are version-sensitive (adaptive vs fixed log-odds threshold); we report the authors’ version.

6 Reproducibility

All code and intermediate results live in `benchmarking/guide-assignment/nonparametric-thresholds/`. Key scripts: `characterize_grna.R` (parameter ranges), `build_comprehensive_sim.R` (simulation), `final_same_cohort.R` (barnyard), `score_comprehensive_panel.R` (panel), `pr_curves.R` + `combine_pr.R` (PR curves). The real geomux (Python `venv .venv_geomux/`) and fishash (R package) were run via `run_geomux*.py` and in-script `fishash()` calls.